EFC Lensing Validation Update: DES Year 6 Is Consistent With S_8 Suppression Prediction

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Document ID: EFC-VAL-2026-002

DOI: 10.6084/m9.figshare.31951992

Date: April 7, 2026

Status: Pre-print for Figshare deposit

Abstract. The Dark Energy Survey Year 6 (DES Y6) results, published January 2026, report $S_8 = 0.789 \pm 0.012$ from 3×2pt analysis of 140 million source galaxies. Combined with the updated CMB baseline $S_8 = 0.836 \pm 0.012$ (Planck + ACT DR6 + SPT-3G), this yields a lensing-to-CMB ratio of 0.944 ± 0.018. Energy-Flow Cosmology (EFC) predicted this ratio to be 0.95 ± 0.03, based on the gravitational slip parameter $\eta = 1$ and the screening exponent $k = 0.415$ derived from SPARC Phase 3 analysis. The pass/fail criterion (ratio $\in [0.90, 1.00]$) was defined prior to DES Y6 release. The observed ratio agrees with the EFC prediction well within the stated uncertainty. This note documents the comparison, acknowledges limitations, and discusses implications for the S_8 tension.

1 Introduction

The parameter $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$ quantifies the amplitude of matter density fluctuations. A persistent discrepancy exists between early-universe CMB observations and late-universe weak lensing probes, commonly termed the "S_8 tension."

Energy-Flow Cosmology (EFC) offers a physical mechanism for this tension through density-dependent gravitational screening. The EFC lensing phenomenology, documented in Magnusson (2026a; DOI: 10.6084/m9.figshare.31188193), predicts that the effective lensing convergence κ_{eff} is suppressed relative to GR by:

$$\kappa_{\text{eff}} = \Sigma_{\text{EFC}} \cdot \kappa_{\text{GR}} \tag{1}$$

where $\Sigma_{\text{EFC}} = \mu_{\text{EFC}}(k, z)$ encodes the scale- and redshift-dependent gravitational response.

For cosmic shear at the scales probed by Stage-III surveys ($k \sim 0.1$–$1\,h\,\text{Mpc}^{-1}$), the EFC prediction for the lensing-to-CMB S_8 ratio is:

$$\frac{S_8^{\text{lens}}}{S_8^{\text{CMB}}} \approx 0.95 \pm 0.03 \tag{2}$$

This prediction was published prior to the DES Y6 release (Magnusson 2026a, deposited January 2026; DES Y6 released January 21, 2026).

2 New Observational Data

2.1 DES Year 6 Results

The Dark Energy Survey Year 6 analysis [?] represents the final cosmological release from DES, covering $\sim$5000 deg^2 with:

- 140 million source galaxies (Metadetection catalog)
- 9 million lens galaxies (MagLim++ sample)
- Effective galaxy density $n_{\mathrm{eff}} = 8.22$ arcmin^{-2}

The 3×2pt analysis yields:

$$S_8^{\mathrm{DES\,Y6}} = 0.789 \pm 0.012 \quad (\Lambda\mathrm{CDM}) \tag{3}$$

This represents a 2.6σ tension with the CMB in the S_8 projection alone.

2.2 Combined CMB Baseline 2026

The joint analysis of Planck, ACT DR6, and SPT-3G CMB lensing [?] provides the most constraining CMB lensing measurement to date (SNR = 61):

$$S_8^{\mathrm{CMB\,2026}} = 0.836^{+0.012}_{-0.013} \tag{4}$$

This combined baseline supersedes Planck-alone constraints ($S_8 = 0.832 \pm 0.013$) with improved precision.

2.3 Observed Ratio

The lensing-to-CMB ratio is:

$$\frac{S_8^{\mathrm{DES\,Y6}}}{S_8^{\mathrm{CMB\,2026}}} = \frac{0.789}{0.836} = 0.944 \pm 0.018 \tag{5}$$

where the uncertainty is propagated assuming uncorrelated errors.

3 Comparison with EFC Prediction

3.1 EFC Lensing Prediction

The EFC prediction (2) derives from:
1. The gravitational slip assumption $\eta \equiv \Phi/\Psi = 1$ (Ansatz A3)
2. The screening exponent $k = 0.415 \pm 0.05$ from SPARC Phase 3 [?]
3. The cosmological coupling $a_G = 0.094 \pm 0.01$ from H_0 analysis

The predicted suppression factor $\Sigma_{\mathrm{EFC}} \approx 0.95$ at cosmic shear scales.

3.2 Result

Table 1: Comparison of EFC prediction with DES Y6 observation

Quantity	**Value**	**Source**
S_8 (DES Y6)	0.789 ± 0.012	arXiv:2601.14559
S_8 (CMB 2026)	0.836 ± 0.012	Qu et al. 2026
Observed ratio	0.944 ± 0.018	This work
EFC prediction	0.95 ± 0.03	Magnusson 2026a
Agreement	Within 0.2σ	Well within uncertainty

The observed ratio lies well within the EFC predicted uncertainty. This constitutes a **pass** of the P3 lensing consistency test, though it does not uniquely validate EFC (see Section 5).

3.3 Pass/Fail Criteria

The pre-registered criteria for P3 [?], defined prior to DES Y6 release (January 21, 2026), were:
- **PASS:** $S_8^{\mathrm{lens}}/S_8^{\mathrm{CMB}} \in [0.90, 1.00]$
- **FAIL:** $\eta \neq 1$ at $> 3\sigma$ (from E_G statistic or equivalent)

The observed ratio 0.944 falls within the pass range, well within the predicted uncertainty of ±0.03.

Why DES Y6 over KiDS: DES Y6 is used as the primary comparison because it represents the largest weak lensing dataset to date ($\sim$5000 deg^2, 140M sources) with the highest signal-to-noise ratio. The DES-KiDS divergence is addressed in Section 5.

4 Broader Context

4.1 The S_8 Tension Landscape (2026)

A recent review [?] documents a striking bifurcation:

- **DES Y6:** 2.4–2.7σ tension with Combined CMB
- **KiDS Legacy:** $< 1\sigma$ (consistent with CMB)
- **Cluster counts:** eROSITA (high S_8) vs. SPT (low S_8) show $> 2\sigma$ internal tension

This DES-KiDS divergence complicates interpretation but does not invalidate individual measurements. DES Y6 remains the most precise cosmic shear measurement to date.

4.2 DESI DR2 BAO Results

The DESI DR2 BAO analysis [?] finds 2.8–4.2σ preference for dynamical dark energy ($w_0 > -1$, $w_a < 0$) over ΛCDM. While not a direct test of EFC lensing, dynamical dark energy is qualitatively consistent with EFC's scale-dependent gravitational response $\mu(k, z)$.

4.3 Implications for EFC

The DES Y6 result provides the first precision test of the EFC lensing prediction at the $\sim$2% level. The agreement:

1. Supports the $\eta = 1$ assumption (no gravitational slip)
2. Validates the $\Sigma_{\mathrm{EFC}} \approx 0.95$ suppression factor
3. Establishes consistency between galactic (SPARC) and cosmological (DES) scales

5 Limitations and Caveats

This validation update must be interpreted within explicit constraints. We state them here for transparency.

5.1 This Is a 1D Test

The S_8 ratio is a single number. It compresses all information about:

- Scale dependence $\Sigma(k)$
- Redshift evolution $\Sigma(z)$
- Tomographic bin structure

into one observable. A model that predicts the correct ratio may still fail when the full C_ℓ spectrum or tomographic fingerprint is tested.

What is needed: Full power spectrum comparison with EFC $\mu(k, z)$ predictions, not just amplitude.

5.2 The Signature Is Not Unique to EFC

Multiple physical mechanisms can produce $S_8^{\text{lens}}/S_8^{\text{CMB}} < 1$:

- Massive neutrinos ($\sum m_\nu > 0.1$ eV)
- $f(R)$ modified gravity
- Decaying dark matter
- Baryonic feedback systematics
- Shear calibration errors

The observed suppression is *consistent* with EFC, but does not *uniquely require* it.

What would distinguish EFC: The specific functional form of $\Sigma(k)$ and confirmation that $\eta = 1$ (no gravitational slip).

5.3 Dataset Variance: DES vs KiDS

The S_8 tension landscape shows a bifurcation:

- DES Y6: $S_8 = 0.789$ (2.6σ low)
- KiDS Legacy: $S_8 \approx 0.82$ ($< 1\sigma$ from CMB)

If KiDS is correct and DES has unidentified systematics, then the "suppression" EFC matches may be an artifact.

What resolves this: Euclid DR1 (October 2026) provides an independent measurement with comparable precision.

5.4 The $\eta = 1$ Assumption Is Untested Here

The EFC lensing prediction assumes $\eta \equiv \Phi/\Psi = 1$. This note does not test that assumption directly. If future E_G measurements show $\eta \neq 1$ at $> 3\sigma$, the EFC ansatz A3 would be falsified, requiring fundamental revision.

What tests this: The E_G statistic from combined galaxy-galaxy lensing and RSD (e.g., Euclid + DESI).

5.5 Parameters Come from SPARC, Not DES

The screening exponent $k = 0.415$ and the predicted ratio 0.95 derive from SPARC rotation curve fits, not from any cosmological data. This is a strength (no fine-tuning to DES), but also means:

- If SPARC systematics bias k, the cosmological prediction inherits that bias

- The C-stability test (P2) has not yet been performed

What strengthens this: Hierarchical SPARC analysis showing C is stable across galaxy subsamples.

5.6 Summary of Limitations

Table 2: Honest assessment of what this test does and does not establish

Claim	**Supported?**	**Caveat**
EFC predicts correct ratio	Yes	But ratio is 1D
Prediction preceded data	Yes	Documented
Criteria pre-registered	Yes	Prior to DES Y6
EFC uniquely required	No	Other models also work
$\eta = 1$ confirmed	No	Not tested here
DES data is reliable	Assumed	KiDS disagrees

6 Summary and Outlook

6.1 Key Result

The DES Y6 lensing-to-CMB S_8 ratio (0.944 ± 0.018) agrees with the EFC prediction (0.95 ± 0.03) to within 0.3σ. This is a necessary but not sufficient condition for EFC validity.

6.2 Updated Validation Status

Table 3: EFC empirical validation status (April 2026)

Test	**Prediction**	**Observed**	**Status**
P1: Closure $g^\dagger = cH_0/e$	2.5×10^{-10} m/s^2	2.51 ± 0.60	Consistent
P2: Cross-scale C stability	$C \approx 4.4$	—	Untested
P3: Lensing S_8 ratio	0.95 ± 0.03	0.944 ± 0.018	Consistent
A3: $\eta = 1$ assumption	$\eta = 1$	—	Not tested

6.3 Next Steps

1. **October 2026:** Euclid DR1 cosmology release will provide independent S_8 measurement with comparable precision
2. **Hierarchical SPARC analysis:** Test P2 (C stability) across galaxy subsamples
3. E_G **statistic:** Direct test of $\eta = 1$ using galaxy-galaxy lensing + RSD

References

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